

Database Schema

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 | 02_Elaboration/Database_Schema.md

Full schema for the Resilio PostgreSQL database hosted on Supabase. Includes table descriptions, column definitions, relationships, Row Level Security policies, and indexing notes.

OVERVIEW

Statistic	Value
Database	PostgreSQL 15 via Supabase
Total tables	20
Authentication	SuperTokens (external); user IDs synced into `users` table
Access control	Row Level Security (RLS) on all tables
Hosting	Supabase managed cloud (data stored outside Nepal — disclosed in privacy notice)

TABLE: users

Central user table. Every registered user has one record here. Role determines which module they access.

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Primary key
email	TEXT	UNIQUE, NOT NULL	User email address
user_role	TEXT	NOT NULL, CHECK IN ('user','therapist','admin')	Application role

Column	Type	Constraints	Description
full_name	TEXT		Display name
avatar_url	TEXT		Cloudinary URL for profile photo
fcm_token	TEXT		Firebase Cloud Messaging device token
is_active	BOOLEAN	DEFAULT true	Account active status
created_at	TIMESTAMPTZ	DEFAULT now()	Account creation timestamp
updated_at	TIMESTAMPTZ	DEFAULT now()	Last profile update

RLS Policies:

- Users can SELECT their own record: `auth.uid() = id`
- Users can UPDATE their own record: `auth.uid() = id`
- Therapists can SELECT customer records for patients with appointments: join-based policy
- Admins have full SELECT and UPDATE access

TABLE: therapist_profiles

Extended profile for users with role = 'therapist'. One-to-one with users.

Column	Type	Constraints	Description
id	UUID	PK	Primary key
user_id	UUID	FK → users.id, UNIQUE	Links to users table
bio	TEXT		Professional biography
specialisations	TEXT[]		Array of specialisation tags
languages	TEXT[]		Languages spoken
consultation_fee	NUMERIC(10,2)	NOT NULL	Fee per session in NPR
is_verified	BOOLEAN	DEFAULT false	Admin verification status
qualification_docs	TEXT[]		Cloudinary URLs of credential documents

Column	Type	Constraints	Description
years_experience	INTEGER		Years of professional experience
rating	NUMERIC(3,2)	DEFAULT 0	Average session rating
total_sessions	INTEGER	DEFAULT 0	Total completed sessions
created_at	TIMESTAMPTZ	DEFAULT now()	

RLS Policies:

- Any authenticated user can SELECT verified therapist profiles: `is_verified = true`
- Therapist can UPDATE their own profile: `auth.uid() = user_id`
- Admins can SELECT all (including unverified) and UPDATE `is_verified`

TABLE: appointments

All therapy appointments (pending, confirmed, completed, cancelled).

Column	Type	Constraints	Description
id	UUID	PK	Primary key
patient_id	UUID	FK → users.id	Patient user
therapist_id	UUID	FK → therapist_profiles.id	Therapist
scheduled_time	TIMESTAMPTZ	NOT NULL	Appointment date and time (UTC)
duration_minutes	INTEGER	DEFAULT 50	Session duration
status	TEXT	CHECK IN ('pending','confirmed','completed','cancelled')	Appointment lifecycle status
payment_status	TEXT	CHECK IN ('pending','paid','refunded','failed')	Payment status
payment_transaction_id	TEXT		eSewa transaction reference
meeting_room_id	TEXT	UNIQUE	WebRTC room identifier
notes	TEXT		Post-session notes (therapist only)

Column	Type	Constraints	Description
created_at	TIMESTAMPTZ	DEFAULT now()	

RLS Policies:

- Patient can SELECT their own appointments: `auth.uid() = patient_id`
- Therapist can SELECT appointments where they are the therapist: `therapist_id IN (SELECT id FROM therapist_profiles WHERE user_id = auth.uid())`
- Only the backend service role can INSERT (via Node.js API)
- Admins have full access

TABLE: subscriptions

Active and historical subscription records per customer.

Column	Type	Constraints	Description
id	UUID	PK	Primary key
user_id	UUID	FK → users.id	Subscriber
plan_id	TEXT	NOT NULL	'FREE', 'SILVER', 'GOLD'
status	TEXT	CHECK IN ('active','expired','cancelled')	Subscription status
payment_method	TEXT		'esewa'
started_at	TIMESTAMPTZ	DEFAULT now()	Subscription start
expires_at	TIMESTAMPTZ		Subscription expiry
transaction_id	TEXT		eSewa transaction reference

Constraint: UNIQUE (user_id) WHERE status = 'active' — prevents duplicate active subscriptions.

RLS Policies:

- User can SELECT their own subscription
- Only service role can INSERT/UPDATE

TABLE: mood_entries

Daily mood logs submitted by customers.

Column	Type	Constraints	Description
id	UUID	PK	
user_id	UUID	FK → users.id	Owner
mood_score	INTEGER	NOT NULL, CHECK (1–5)	Numeric mood rating
mood_label	TEXT		'Very Sad', 'Sad', 'Neutral', 'Happy', 'Very Happy'
note	TEXT		Free-text journal note (max 500 chars)
logged_at	TIMESTAMPTZ	DEFAULT now()	Log timestamp
synced_from_offline	BOOLEAN	DEFAULT false	True if entry was created offline and synced

RLS Policies:

- User can SELECT, INSERT, UPDATE their own entries: `auth.uid() = user_id`
- Therapist can SELECT entries for their patients (via appointment relationship)

TABLE: questionnaire_responses

PHQ-9 and GAD-7 submission records.

Column	Type	Constraints	Description
id	UUID	PK	
user_id	UUID	FK → users.id	
questionnaire_type	TEXT	CHECK IN ('PHQ9','GAD7')	
answers	INTEGER[]	NOT NULL	Array of individual answers
total_score	INTEGER	NOT NULL	Sum of answers

Column	Type	Constraints	Description
severity_band	TEXT		Minimal / Mild / Moderate / Moderately Severe / Severe
crisis_alert	BOOLEAN	DEFAULT false	True if PHQ-9 score $\geq$ 20
completed_at	TIMESTAMPTZ	DEFAULT now()	

RLS Policies:

- User can SELECT and INSERT their own responses
- Therapist can SELECT their patients' responses

TABLE: game_sessions

Records of completed wellness game sessions.

Column	Type	Constraints	Description
id	UUID	PK	
user_id	UUID	FK → users.id	
game_type	TEXT	CHECK IN ('breathing','trivia','affirmation','story')	
score	INTEGER	DEFAULT 0	Game score (0 for breathing)
duration_seconds	INTEGER		Session length
completed_at	TIMESTAMPTZ	DEFAULT now()	

RLS Policies:

- User can SELECT and INSERT their own sessions

TABLE: achievements

Master list of all available achievement types.

Column	Type	Description
id	TEXT	PK — achievement code (e.g. 'FIRST_BREATH')
title	TEXT	Display name
description	TEXT	Achievement description
icon_url	TEXT	Badge icon Cloudinary URL
category	TEXT	'games', 'mood', 'booking', 'questionnaire'

TABLE: achievement_entities (Junction)

Records which achievements each user has unlocked.

Column	Type	Description
id	UUID	PK
user_id	UUID	FK → users.id
achievement_id	TEXT	FK → achievements.id
unlocked_at	TIMESTAMPTZ	

Constraint: UNIQUE (user_id, achievement_id)

TABLE: content (Wellness Content Hub)

All published and pending wellness content items.

Column	Type	Constraints	Description
id	UUID	PK	
author_id	UUID	FK → users.id	Therapist author
title	TEXT	NOT NULL	Content title
body	TEXT		Full article text (for articles)

Column	Type	Constraints	Description
content_type	TEXT	CHECK IN ('article','audio','video','short')	
media_url	TEXT		Cloudinary URL for audio/video
thumbnail_url	TEXT		Cloudinary thumbnail
category	TEXT		e.g. 'anxiety', 'depression', 'sleep'
is_premium	BOOLEAN	DEFAULT false	Requires active subscription
status	TEXT	CHECK IN ('draft','pending_review','published','rejected')	
published_at	TIMESTAMPTZ		Set when admin approves
created_at	TIMESTAMPTZ	DEFAULT now()	

RLS Policies:

- Public: SELECT WHERE status = 'published'
- Premium content: SELECT WHERE status = 'published' AND (NOT is_premium OR subscription check)
- Therapist: SELECT/INSERT/UPDATE their own content
- Admin: Full access

TABLE: categories

Tag taxonomy for content classification.

Column	Type	Description
id	UUID	PK
name	TEXT	Category name (e.g. 'Anxiety', 'Sleep')
icon_url	TEXT	Category icon

TABLE: favorites

User's saved content items and therapist profiles.

Column	Type	Description
id	UUID	PK
user_id	UUID	FK → users.id
item_id	UUID	ID of the favorited item
item_type	TEXT	'content' or 'therapist'
saved_at	TIMESTAMPTZ	

Constraint: UNIQUE (user_id, item_id, item_type)

TABLE: notifications

In-app notification records.

Column	Type	Description
id	UUID	PK
user_id	UUID	FK → users.id
title	TEXT	Notification title
body	TEXT	Notification body
type	TEXT	'booking_confirmed', 'session_reminder', 'content_approved', etc.
is_read	BOOLEAN	DEFAULT false
created_at	TIMESTAMPTZ	

TABLE: transactions

Financial transaction audit log.

Column	Type	Description
id	UUID	PK
user_id	UUID	Payer
appointment_id	UUID	FK → appointments.id (nullable for subscriptions)
subscription_id	UUID	FK → subscriptions.id (nullable for appointments)
amount	NUMERIC(10,2)	NPR amount
payment_method	TEXT	'esewa'
transaction_ref	TEXT	eSewa transaction reference
status	TEXT	'success', 'failed', 'refunded'
created_at	TIMESTAMPTZ	

TABLE: preferences

User's onboarding preference selections.

Column	Type	Description
id	UUID	PK
user_id	UUID	FK → users.id, UNIQUE
goals	TEXT[]	Selected wellness goals
concerns	TEXT[]	Areas of concern (anxiety, stress, etc.)
preferred_language	TEXT	'en' or 'ne'
notification_enabled	BOOLEAN	
updated_at	TIMESTAMPTZ	

TABLE: quotes

Daily motivational quotes displayed on home screen.

Column	Type	Description
id	UUID	PK
text	TEXT	Quote body
author	TEXT	Quote attribution
category	TEXT	Thematic tag

TABLE: tips

Wellness tip cards displayed in content hub.

Column	Type	Description
id	UUID	PK
title	TEXT	Tip title
body	TEXT	Tip content
category	TEXT	

ROW LEVEL SECURITY — SUMMARY

Table	Customer	Therapist	Admin	Backend (service role)
users	Own record only	Own + patients	All	All
therapist_profiles	Verified only	Own profile	All	All
appointments	Own appointments	Own patients'	All	All
mood_entries	Own entries	Own patients'	None	All
questionnaire_responses	Own responses	Own patients'	None	All
content	Published only	Own + published	All	All

Table	Customer	Therapist	Admin	Backend (service role)
subscriptions	Own	None	All	All
notifications	Own	Own	All	All
transactions	Own	None	All	All

KEY INDEXES

Table	Column	Index Type	Reason
mood_entries	user_id, logged_at	Composite B-tree	Fast 7/30-day range queries
appointments	patient_id, scheduled_time	Composite B-tree	Calendar and upcoming session queries
appointments	therapist_id, status	Composite B-tree	Therapist appointment list filtering
content	status, content_type	Composite B-tree	Published content hub browsing
questionnaire_responses	user_id, questionnaire_type	Composite B-tree	History queries
therapist_profiles	is_verified	Partial B-tree	Directory listing (only verified)